DATABASE

INPUT DATA AND PRE-PROCESSING

Satellite reflectance	Bathymetric survey	Tide-model covariate	Open bathymetry
HLS v2.0 S30 + L30	XYZ soundings local port zeros	GOT5.6 + pyTMD harmonic tide	EMODnet + GEBCO
Cloud / invalid-pixel screening	Transform XYZ to local UTM	Predict tide at 10 min steps	Reproject / inspect source coverage and product metadata
Harmonize B02/B03/B04 + NIR	Review exact XY duplicates	Sample each scene overpass	CONTEXT MAPS ONLY Not training truth No datum substitution
Quality-weighted composite · ±30 d	Median depth per 30 m cell	Subtract survey-day mean	
FOUR-BAND COMPOSITE	30 m DEPTH CELLS	RELATIVE TIDE ANOMALY	SOURCE CONTEXT

COMMON 30 m PIXEL OVERLAY

HLS reflectance + surveyed depth + weighted tide anomaly; retain each local port zero

BATHYMETRY ESTIMATORS

Stumpf log-ratio · Random Forest · HistGradientBoosting · RBF-SVR

SPATIAL HOLD-OUT EVALUATION

5 grouped folds · 500 m blocks · 90 m buffer
Out-of-fold RMSE, MAE, bias and depth-stratified errors
Block-bootstrap intervals and sensitivity analyses

SUPPORTED FULL-DATA MAPS

Random Forest · within 90 m of survey support



Accuracy is reported from held-out folds; fitted maps visualise spatial patterns only.